

CURRICULUM VITAE

STEPHANIE VARGAS AGUILAR

Department of Molecular Biology
UT Southwestern Medical Center
Dallas, United States
stephanie.vargasaguilar@utsouthwestern.edu

EDUCATION

- July 2018 **Ph.D in Biology and Immunology**
Humboldt University of Berlin, Germany
Dissertation: „Investigation of the molecular mechanisms that enable the self-renewal of tissue-resident macrophages”
Honors: *magda cum laude*
Advisor: Dr. Michael Sieweke
- May 2012 **German Diploma (B.A. and M.S.) in Molecular Medicine**
Albert Ludwig University of Freiburg, Germany
Diploma Thesis: „Characterization of mutant β -catenin-variants in β -catenin^{-/-} mouse embryonic stem cells”
Advisor: Dr. Rolf Kemler

ACADEMIC POSITIONS

- 2020 – present **Postdoctoral fellow with Dr. Eric Olson**
Field of study: Immunology and heart regeneration
UT Southwestern Medical Center, Dallas, United States
- 2013 – 2018 **Graduate Student with Prof. Dr. Michael Sieweke**
Field of study: Molecular mechanisms of macrophage aging
Centre d'Immunologie de Marseille-Luminy, France
Max Delbrück Center for Molecular Medicine Berlin, Germany
- 2011 – 2012 **Undergraduate Research Associate with Dr. Rolf Kemler**
Field of study: Early mammalian development
Max Planck Institute for Immunobiology and Epigenetics, Freiburg
- 2008 – 2010 **Research Assistant with Dr. Thomas Reinheckel**
Field of study: In Vivo Functions of Proteases
Albert Ludwig University of Freiburg, Germany
Institute for Molecular Medicine and Cell Research

PEER-REVIEWED PUBLICATIONS

Vargas Aguilar, S, Cui, M, Tan, W, Sanchez-Ortiz, E, Bassel-Duby, R, Liu, N, and Olson, EN. The PD-1 – PD-L1 pathway maintains an immunosuppressive environment essential for neonatal heart regeneration. Nat. Cardiovasc. Res. 2024 Mar; 10.1038/s44161-024-00447-7.

Gainullina, A, Sergushichev A, Artyomov MH, **ImmGen Consortium**. Resource Network analysis of large-scale ImmGen and Tabula Muris datasets highlights metabolic diversity of tissue mononuclear phagocytes. Cell Reports. 2023 Feb 28;42(2):112046. PMID: 36708514.

Subramanian S, Busch CJ, Molawi K, Geirsdottir L, Maurizio J, **Vargas Aguilar S**, Belahbib H, Gimenez G, Yuda RAA, Burkon M, Favret J, Gholamhosseinian Najjar S, de Laval B, Kandalla PK, Sarrazin S, Alexopoulou L, Sieweke MH. Long-term culture-expanded alveolar macrophages restore their full epigenetic identity after transfer in vivo. Nature Immunology. 2022 Mar;23(3):458-468. PMID: 35210623.

Vargas Aguilar S. (as part of the ImmGen Consortium) The Immunological Genome Project. ImmGen at 15. Nature Immunology. 2020 Jul;21(7):700-703. PMID: 32577013

Imperatore F, Maurizio J, **Vargas Aguilar S**, Busch CJ, Favret J, Kowenz-Leutz E, Cathou W, Gentek R, Perrin P, Leutz A, Berruyer C, Sieweke MH. SIRT1 regulates macrophage self-renewal. EMBO Journal. 2017 Aug 15;36(16):2353-2372. PMID: 28701484

Matcovitch-Natan O, Winter DR, Giladi A, **Vargas Aguilar S**, Spinrad A, Sarrazin S, Ben-Yehuda H, David E, Zelada González F, Perrin P, Keren-Shaul H, Gury M, Lara-Astaiso D, Thaïss CA, Cohen M, Bahar Halpern K, Baruch K, Deczkowska A, Lorenzo-Vivas E, Itzkovitz S, Elinav E, Sieweke MH, Schwartz M, Amit I. Microglia development follows a stepwise program to regulate brain homeostasis. Science. 2016 Aug 19;353(6301):aad8670. PMID: 27338705.

Dennemärker J, Lohmüller T, Müller S, **Vargas Aguilar S**, Tobin DJ, Peters C, Reinheckel T. Impaired turnover of autophagolysosomes in cathepsin L deficiency. Biological Chemistry. 2010 Aug;391(8):913-22. PMID: 20536383.

GRANTS

2025 – present	Career Development Award American Heart Association (\$231,000.00)
2022 - 2024	Postdoctoral Fellowship American Heart Association (\$145,020.00)
2016 - 2017	PhD Scholarship Aides doctorales en France Foundation ARC Pour La Recherche Sur Le Cancer (€ 49,300)
2013 - 2016	PhD Scholarship International PhD Programme of the Helmholtz Graduate School (€ 79,457.18)

PRESENTATIONS

Oral Presentations

2023	<u>Differential T cell usage during neonatal cardiac injury</u> , NPAW Symposium 2023. UT Southwestern Medical Center, Dallas, USA.
2022	<u>El rol del sistema inmune en la regeneración cardíaca</u> , Charlas Virtuales CISEAL. Center for Research on Health in Latin America, Quito, Ecuador.
2022	<u>The PD-1/PD-L1 pathway provides an immunosuppressive environment necessary for mouse heart regeneration</u> . EMBO Workshop on the molecular and cellular basis of regeneration and tissue repair, Barcelona, Spain
2020	<u>A PIWI protein protects alveolar macrophages from aging-associated decline</u> . Immunity in Regeneration, Kick-off Mini Symposium, Center for Regenerative Therapies, Dresden, Germany.
2019	<u>A PIWI protein controls the replicative lifespan of alveolar macrophages</u> . 33rd Annual Conference of the European Macrophage and Dendritic Cell Society, Marseille, France.
2015	<u>The PIWI/piRNA pathway in macrophage self-renewal</u> . Scientific Retreat of the Centre d'Immunologie Marseille Luminy, Marseille, France.

Poster Presentations

- 2023 Vargas Aguilar S, Cui M, Tan W, Liu N, Bassel-Duby R, Olson EN. The PD1/PD-L1 pathway maintains an immunosuppressive environment essential for neonatal heart regeneration. BCVS Scientific Sessions from the American Heart Association, Boston, USA.
- 2022 Vargas Aguilar S, Cui M, Tan W, Liu N, Bassel-Duby R, Olson EN. The PD1/PD-L1 pathway provides an immunosuppressive environment necessary for mouse heart regeneration. NPAW Symposium 2022, UT Southwestern Medical Center, Dallas, USA.
- 2018 Vargas Aguilar S, Subramanian S, Singh M, Gimenez G, Favret J, Mossadegh-Keller N, Huang H, Izsvák Zsuzsanna, Sieweke MH. A small isoform of Piwil2 controls the proliferation and lifespan of murine macrophages. EMBO Workshop on piRNAs and PIWI proteins, Montpellier, France.
- 2017 Vargas Aguilar S, Gimenez G, Subramanian S, Mossadegh-Keller N, Sieweke MH. Novel somatic function of Piwil2 in mouse macrophage self-renewal. Nature Conference on Regeneration, Milan, Italy.
- 2016 Vargas Aguilar S, Sieweke MH. Transcriptional control of microglia proliferation and activation in the CNS. NeuroMac, Freiburg, Germany.
- 2016 Vargas Aguilar S, Gimenez G, Sieweke MH. A short isoform of Piwil2 is required for the self-renewal of murine macrophages. EMBO Workshop on piRNAs and PIWI proteins, Montpellier, France.

PROFESSIONAL ACTIVITIES

Service

Chair of the Webinar Committee of the International Society for Regenerative Biology, Zone of the Americas (July 2021 – present)
Vice-president of the UT Southwestern Postdoctoral Association (January 2023 – January 2024)

Memberships

American Heart Association
International Society for Heart Research
American Association of Immunologists
Society for Developmental Biology
International Society for Regenerative Biology
Latin American Society for Developmental Biology
Organization for Women in Science for the Developing World

SCIENTIFIC OUTREACH

- 2023 My personal and scientific path as a minority in STEM
Presentation as an invited speaker for the Hispanic Heritage Month Symposium 2023, UT Southwestern Medical Center, Dallas, USA.
- 2022 El anhelo de volver al corazón infantil
Presentation (virtual) as an invited speaker for the Science Outreach Association “Villaviciosa ConCiencia” , Villaviciosa, Spain.

LANGUAGES

Spanish (Mother tongue)
English (Fluently spoken and written)
German (Fluently spoken and written)
French (Proficient)